Notes: Dong, Fisman, Wang & Xu (JFE, 2021)

Air pollution, affect, and forecasting bias: Evidence from Chinese financial analysts

Contents

1 Big picture

- **Research question.** Do transitory environmental conditions affect the judgments of expert financial intermediaries? The paper studies whether local air pollution during corporate site visits affects sell-side analysts' subsequent EPS forecasts.
- **Core setting.** Chinese listed firms disclose corporate site visits. The authors match site-visit dates and cities to daily air-quality data and to analyst earnings forecasts issued shortly after the visit.
- **Main result.** Worse air quality at the visit city on the visit date is associated with lower subsequent forecast optimism. In the fully controlled baseline, the coefficient on scaled AQI is about -3.77.
- **Magnitude.** Moving from "good/excellent" air quality to severe pollution implies roughly a one percentage point decline in forecast optimism, measured relative to forecast-price scaling.
- **Mechanism.** The evidence points toward an affect or mood channel rather than pure information about firm fundamentals: the effect is strongest for forecasts issued soon after the visit and fades with delay.
- **Not simply firm information.** The effect is weaker, not stronger, for firms in high-pollution industries, which argues against the view that analysts infer bad fundamentals from a firm's own emissions.
- **Debiasing.** The pollution effect is attenuated when analysts from other brokerages are present at the same visit, consistent with cross-brokerage disagreement or "wisdom of crowds" reducing affect-driven bias.
- **Economic takeaway.** Even sophisticated experts making high-stakes forecasts appear influenced by irrelevant environmental conditions. The paper therefore links behavioral finance, affect, pollution, and information production by financial analysts.

2 Incremental contribution of the paper

- **From retail investors to expert agents.** Prior work links weather and pollution to investor mood, trading, and stock returns. This paper shifts the focus to professional analysts who are trained, specialized, and financially incentivized to form accurate beliefs.
- **A cleaner event setting.** The corporate site-visit disclosure regime in China gives the authors a precise time and location at which analysts are exposed to air pollution before issuing forecasts. This is sharper than studying ambient pollution around a trading desk or city-wide investor sentiment.
- **Forecast-level outcome.** The paper studies analyst EPS forecast bias rather than market prices alone. This lets the authors observe a concrete belief or judgment produced after the environmental exposure.
- **Affect versus information.** The paper does more than show a pollution-forecast correlation. It uses timing, pollution persistence, high-pollution industries, analyst home-city adaptation, analyst ability, and group visits to separate affect from alternative information channels.
- **Debiasing mechanism.** The cross-brokerage group-visit evidence suggests that social aggregation can reduce individual affective bias. This is an incremental insight beyond documenting that pollution worsens forecasts.
- **Methodological contribution.** The paper shows how site-visit records can be used as a micro event study of expert belief formation. The design can be extended to other contexts in which analysts, investors, inspectors, or regulators visit firms.

3 Data and measurement

3.1 Corporate site-visit data

The paper hand-collects corporate site-visit disclosures for firms listed on the Shenzhen Stock Exchange from 2009 to 2015. These disclosures report the date, city, firm, and visitor identities for corporate site visits. The raw set contains more than twenty thousand disclosures and includes visitors such as sell-side analysts, investors, fund managers, and reporters. The empirical analysis focuses on sell-side analysts who issue EPS forecasts after a visit.

Interpretation: The institutional feature that matters is timing and location. Because visits are disclosed with exact dates and cities, the authors can attach a local daily pollution exposure to the analyst's information-gathering event.

3.2 Forecast sample

The main sample keeps analyst-firm-visit observations in which a sell-side analyst releases an EPS forecast within a short window after the site visit. The baseline window is forecasts issued within 15 calendar days after the visit; robustness exercises use shorter and longer windows.

The main outcome is forecast optimism:

$$\text{ForecastOptimism}_{ijt} = 100 \times \frac{\widehat{EPS}_{ijt} - EPS^{realized}_{jt}}{P_{j,t^-}},$$

where $\widehat{EPS}_{ijt}$ is analyst i's forecast for firm j, $EPS^{realized}_{jt}$ is realized EPS, and P_{j,t^-} is the stock price before the forecast.

Interpretation: A higher value means the analyst is more optimistic relative to realized earnings. The paper studies whether the analyst becomes less optimistic after visiting a firm on a polluted day.

3.3 Pollution exposure

The main explanatory variable is the Air Quality Index in the visit city on the visit date:

$$\text{AQI}_{c,t}.$$

The variable is scaled by 1000 in the regressions. The AQI/API series summarizes local air pollution using official environmental monitoring data. The paper also uses air-quality categories and lag/lead values of pollution to test robustness and timing.

Interpretation: The site-city and site-date pollution measure is the treatment-like variable. It is intended to capture the analyst's ambient environmental condition while gathering information at the firm.

3.4 Weather and firm controls

The paper controls for local weather conditions and firm characteristics:

- hours of sunshine, temperature, humidity, precipitation, and wind speed;
- firm size, market-to-book, intangible assets, return volatility, turnover, and prior stock return;
- analyst attention, number of firms followed, and number of forecasts issued.

Interpretation: Weather controls are important because prior behavioral-finance work documents direct weather effects on mood and forecasts. Firm controls reduce the possibility that pollution is proxying for firm size, risk, growth opportunities, or investor attention.

3.5 Fixed effects and sample size

The main regressions include combinations of year-quarter fixed effects, day-of-week fixed effects, industry fixed effects, city fixed effects, and analyst fixed effects. The baseline forecast-level sample in the main tables contains 3,824 observations.

Interpretation: Analyst fixed effects compare an analyst to herself across visits. City fixed effects compare more and less polluted days within a city. These fixed effects make the design closer to using daily pollution fluctuations around site visits rather than persistent cross-city or cross-analyst differences.

4 Model and empirical specifications

4.1 Baseline forecast-bias regression

The main empirical specification is:

$$\text{ForecastOptimism}_{ijv} = \alpha + \beta \text{AQI}_{c(v),t(v)} + \Gamma' X_{ijv} + \mu_q + \mu_{dow} + \mu_{industry} + \mu_{city} + \mu_i + \varepsilon_{ijv}.$$

Here, v indexes a site visit, $c(v)$ is the visit city, $t(v)$ is the visit date, and i indexes analysts. X_{ijv} includes weather and firm/analyst controls.

Interpretation: The coefficient β measures how forecast optimism changes with visit-day pollution, holding constant analyst, city, industry, calendar, weather, and firm characteristics. A negative coefficient means polluted visit days lead to more pessimistic forecasts.

4.2 AQI category specification

To avoid relying only on a linear AQI specification, the paper also uses pollution-category indicators:

$$\text{ForecastOptimism}_{ijv} = \sum_k \beta_k \mathbf{1}\{\text{AQI} \in k\} + Controls + FE + \varepsilon_{ijv},$$

where the least-polluted category is omitted.

Interpretation: Monotonicity across AQI categories is important. If more severe pollution categories have increasingly negative coefficients, this supports a dose-response interpretation.

4.3 Persistence and placebo timing

The paper tests whether pollution on days near the visit matters:

$$\text{ForecastOptimism}_{ijv} = \beta_0 \text{AQI}_t + \beta_{-k} \text{AQI}_{t-k} + \beta_{+k} \text{AQI}_{t+k} + Controls + FE + \varepsilon_{ijv}.$$

Interpretation: If visit-day AQI predicts forecasts but lagged and lead AQI do not, the effect is less likely to be driven by persistent city conditions or omitted local trends.

4.4 Forecast delay and horizon

The paper studies whether the effect fades when more days pass between the visit and the forecast. It also estimates specifications with forecast horizon interactions:

$$\text{ForecastOptimism}_{ijv} = \beta_1 \text{AQI}_v + \beta_2 \log(\text{Horizon}_{ijv}) + \beta_3 \text{AQI}_v \times \log(\text{Horizon}_{ijv}) + Controls + FE + \varepsilon_{ijv}.$$

Interpretation: Delay from visit to forecast is used to test whether the mood effect dissipates. Forecast horizon asks whether forecasts about more distant earnings, which are generally less precise, are more sensitive to environmental mood.

4.5 Heterogeneity tests

The paper estimates interaction specifications for firm type, analyst adaptation, analyst skill, and group visits:

$$\text{ForecastOptimism} = \beta_1 \text{AQI} + \beta_2 Z + \beta_3 \text{AQI} \times Z + Controls + FE + \varepsilon.$$

The interacting variable Z includes high-pollution-industry status, analyst home-city pollution surprise, analyst experience/skill, and whether other analysts attended the same visit.

Interpretation: These tests are designed to discriminate mechanisms. If pollution captures information about the firm's own emissions, the effect should be stronger for high-pollution firms. If it captures affect, it should fade with delay and be mitigated by social aggregation or adaptation.

5 Identification and empirical design

5.1 Core identification idea

The design relies on short-run variation in local air quality at the site-visit city on the date of the visit. Conditional on fixed effects, weather controls, and firm/analyst controls, the paper compares forecasts issued after more polluted versus less polluted visits.

Interpretation: The identifying variation is not a randomized experiment, but it is high-frequency and localized. The key assumption is that daily AQI at the visit city is not systematically correlated with unobserved information that analysts learn about the firm and that also predicts forecast optimism.

5.2 Why site visits help identification

Site visits create a moment when analysts are physically exposed to local pollution while gathering firm-specific information. The visit-date structure is crucial because it links pollution exposure to a particular decision event and subsequent forecast.

Interpretation: This is stronger than comparing analysts in polluted versus clean cities generally. The same analyst may visit different firms on different days, and the same city has cleaner and dirtier days.

5.3 Fixed effects logic

The main specification layers fixed effects:

- year-quarter fixed effects absorb macroeconomic and reporting-period conditions;

- day-of-week fixed effects absorb systematic weekday patterns in visits and forecasts;
- industry fixed effects absorb sector-level differences in forecastability and pollution exposure;
- city fixed effects absorb persistent city-level pollution, development, and analyst-market differences;
- analyst fixed effects absorb time-invariant analyst optimism and ability.

Interpretation: With analyst and city fixed effects, identification is driven by within-analyst and within-city fluctuations in pollution around visits, rather than by stable differences between analysts or cities.

5.4 Main threats and how the paper addresses them

- **Weather confounding.** Weather affects mood and may correlate with pollution. The paper controls for sunshine, temperature, humidity, precipitation, and wind speed.
- **Firm emissions information.** Analysts might infer negative fundamentals from pollution if the visited firm is itself an emitter. The high-pollution-industry test shows the pollution effect is not stronger for such firms.
- **Persistent city conditions.** The lead/lag AQI tests show that visit-day pollution matters more than pollution on neighboring days.
- **Forecast timing.** The forecast-delay evidence shows that the effect attenuates as more time passes after the visit, consistent with a short-lived affect channel.
- **Analyst heterogeneity.** Skill, experience, and star status do not fully explain the relation. Analyst fixed effects absorb stable analyst-level traits.
- **Social debiasing.** Cross-brokerage group visits attenuate the relation, suggesting that discussion or diversity of views reduces affective bias.

Interpretation: The paper's credibility comes from the joint pattern. No single table proves causality, but the baseline, dose-response, lead/lag, delay, high-pollution-industry, and group-visit results all point toward a pollution-induced affect channel.

5.5 Interpreting the estimates

A negative β in the baseline regression means that a higher AQI on the visit date predicts lower subsequent forecast optimism. Because the dependent variable is forecast EPS minus realized EPS scaled by price and multiplied by 100, the coefficient can be interpreted as a percentage-point change in forecast optimism.

Interpretation: The estimates should be read as evidence that transitory environmental conditions shift analyst forecasts in a pessimistic direction. The results do not imply that pollution permanently changes analyst beliefs; indeed the delay evidence suggests the effect is temporary.

6 Tables and figures

6.1 Table 1: Summary statistics; Table 2: The relation between air pollution and analyst forecast optimism

Read:

- Table 1 defines the main variables and reports forecast-level and firm-year summary statistics.

- Mean forecast optimism is 2.051, indicating that analysts are optimistic on average.
- Mean AQI is 0.089 after scaling by 1000, with substantial variation across visits.
- Table 2 is the baseline result. The AQI coefficient is negative in all columns and remains significant after adding fixed effects and controls.
- The fully controlled coefficient is -3.769, implying that dirtier air on the visit day predicts lower forecast optimism.

Detailed interpretation: Table 1 establishes the empirical setting: the paper studies a forecast-level panel with meaningful variation in both forecast errors and air quality. Table 2 shows that the core relation is not absorbed by city, industry, analyst, or calendar controls. These two tables provide the foundation for the paper. The rest of the paper asks whether the baseline relation is truly about pollution-induced affect rather than other information or selection channels.

Table 1
Summary statistics.

Forecast_Optimism denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the Air Quality Index of the site visit city on the visit date, scaled by 1000. log(*Horizon*) denotes the natural logarithm of the days between the forecast date and the corresponding date of the actual earnings announcement. *Hours_of_Sun* denotes hours of sun of the site visit city on the visit date (0.1h). *Temperature* denotes the average temperature of the site visit city on the visit date (0.1°C). *Humidity* denotes the average humidity of the site visit city on the visit date (1%). *Precipitation* denotes the total precipitation of the site visit city on the visit date (0.1mm). *Wind_Speed* denotes the average wind speed of the site visit city on the visit date (0.1m/s). log(*Assets*) denotes the natural logarithm of total assets at the beginning of the year when the site visit took place (visit year). *Market_to_Book* denotes the ratio of market value of equity to book value of equity at the beginning of the visit year. *Intangible_Asset* denotes the ratio of intangible assets to total assets at the beginning of the visit year. *Volatility* denotes daily volatility of stock returns during the year prior to the visit year. *Turnover* denotes the daily turnover rate of the visit year. *Return* denotes annual stock returns of the year prior to the visit year. *Analyst_Attention* denotes the natural logarithm of the number of analysts following the firm during the visit year. *Follow_Co_Num* denotes the natural logarithm of the number of companies the analyst followed during the visit year. *Forecast_Num* denotes the natural logarithm of the number of reports issued by the analyst during the visit year. Panel A provides summary statistics based on the main sample of forecast × analyst visit observations. Panel B provides summary statistics collapsed to the firm-year level.

Panel A: Sample for main analysis			
Variable name	Mean	StdDev	Observations
Forecast_Optimism	2.051	3.486	3824
AQI	0.089	0.052	3824
log(*Horizon*)	5.920	0.828	3824
Hours_of_Sun	49.978	41.051	3824
Temperature	172.825	91.507	3824
Humidity	68.855	17.038	3824
Precipitation	37.127	109.890	3824
Wind_Speed	22.201	10.037	3824
Panel B: firm-year aggregates			
Variable name	Mean	StdDev	Observations
log(*Assets*)	21.740	1.054	1046
Market_to_Book	3.124	1.743	1046
Intangible_Asset	0.045	0.050	1046
Volatility	0.028	0.006	1046
Turnover	2.787	2.163	1046
Return	0.253	0.615	1046
Analyst_Attention	2.428	0.755	1046
Follow_Co_Num	2.328	0.802	1046
Forecast_Num	2.867	1.049	1046

Table 1: Summary statistics. The cropped table reports the sample structure, main variables, and forecast/firm summary statistics used throughout the paper.

Table 2
The relation between air pollution and analyst forecast optimism.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the Air Quality Index of the visit city on the visit day, scaled by 1000. Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature, Humidity, Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility, Turnover, Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Appendix D shows the results including point estimates for all control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)
Dependent variable		*Forecast_Optimism*		
AQI	-3.558^{***}	-2.129^{*}	-4.206^{***}	-3.769^{***}
	(1.072)	(1.104)	(1.322)	(1.420)
Year-quarter FEs		Yes	Yes	Yes
Day of week FEs		Yes	Yes	Yes
Industry FEs			Yes	Yes
City FEs			Yes	Yes
Analyst FEs			Yes	Yes
Controls				Yes
Observations	3824	3824	3824	3824
R-squared	0.004	0.065	0.443	0.608

Table 2: Baseline AQI-forecast relation. The table shows the negative relation between site-day AQI and subsequent forecast optimism under increasingly rich fixed effects and controls.

6.2 Table 3: The effect of different AQI categories; Table 4: The effect of pollution persistence on forecast optimism

Read:

- Table 3 replaces linear AQI with AQI categories. The coefficients generally become more negative as pollution categories worsen.
- In the full specification, the most polluted category, AQI above 300, has a coefficient around -1.190 relative to the cleanest category.

- Table 4 adds AQI on prior and future days around the site visit.
- Visit-day AQI remains negative and significant, while AQI before and after the visit is generally insignificant.

Detailed interpretation: Table 3 is a dose-response check. It shows that the result is not driven by assuming a linear AQI effect. Table 4 is a timing placebo. It shows that the relevant pollution is pollution when the analyst is actually visiting, not persistent local pollution around the visit. Together, these tables support the interpretation that the site-day environmental exposure matters for the analyst's subsequent judgment.

Table 3

The effect of different AQI categories.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. $AQI50-100$, $AQI100-150$, $AQI150-200$, $AQI200-300$, and $AQI300+$ are indicator variables that correspond to each of the government's air pollution categories ($AQI < 50$ is the omitted category). See the text for details. Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature, Humidity, Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility, Turnover, Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)
Dependent variable		*Forecast_Optimism*		
$AQI50-100$	−0.006	0.099	−0.232	−0.376
	(0.152)	(0.150)	(0.221)	(0.230)
$AQI100-150$	−0.342*	−0.161	−0.567**	−0.664**
	(0.181)	(0.191)	(0.256)	(0.262)
$AQI150-200$	−0.443**	−0.255	−0.779***	−0.856***
	(0.223)	(0.215)	(0.269)	(0.297)
$AQI200-300$	−0.567*	−0.296	−1.228***	−1.062***
	(0.293)	(0.287)	(0.366)	(0.371)
$AQI300+$	−1.522***	−0.988***	−1.057*	−1.190*
	(0.289)	(0.340)	(0.578)	(0.624)
Year-quarter FEs		Yes	Yes	Yes
Day of week FEs		Yes	Yes	Yes
Industry FEs			Yes	Yes
City FEs			Yes	Yes
Analyst FEs			Yes	Yes
Controls				Yes
Observations	3824	3824	3824	3824
R-squared	0.006	0.067	0.444	0.609

Table 3: AQI categories. More severe pollution categories are associated with increasingly lower optimism relative to the omitted clean-air category.

Table 4

The effect of pollution persistence on forecast optimism.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the Air Quality Index of the visit city on the visit day, scaled by 1000. *AQI_Past*5, *AQI_Past*7, and *AQI_Past*10 denote AQI of the site visit city 5, 7, and 10 days prior to the visit date respectively, scaled by 1000. *AQI_Forward*5, *AQI_Forward*7, and *AQI_Forward*10 denote AQI of the site visit city 5, 7, and 10 days following the visit date, respectively, scaled by 1000. Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature, Humidity, Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility, Turnover, Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable			*Forecast_Optimism*			
AQI	−3.548**	−3.813***	−3.772***	−3.782***	−3.841***	−3.540**
	(1.481)	(1.443)	(1.420)	(1.407)	(1.421)	(1.461)
*AQI_Past*5	−1.501					
	(1.649)					
*AQI_Past*7		0.378				
		(1.346)				
*AQI_Past*10			−0.360			
			(1.474)			
*AQI_Forward*5				0.181		
				(1.764)		
*AQI_Forward*7					0.501	
					(1.374)	
*AQI_Forward*10						−1.410
						(1.142)
Year-quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes
Day of week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes
City FEs	Yes	Yes	Yes	Yes	Yes	Yes
Analyst FEs	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3824	3822	3822	3824	3824	3824
R-squared	0.609	0.608	0.608	0.608	0.608	0.609

Table 4: Pollution persistence. Visit-day AQI is predictive; past and future AQI are not, which weakens omitted-trend explanations.

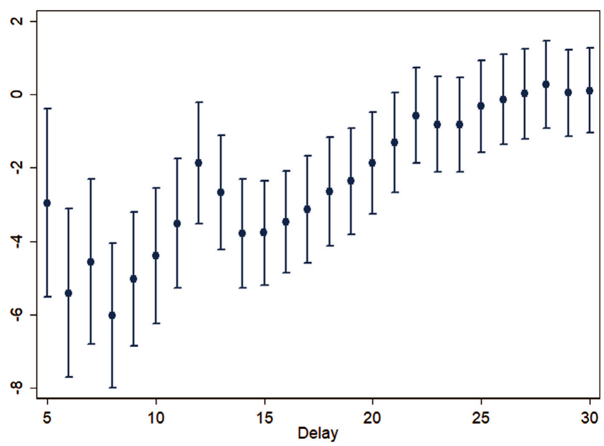
6.3 Figure 1: The attenuating effect of forecast delay; Table 5: The effect of firm type

Read:

- Figure 1 plots AQI coefficients for increasingly long forecast windows after the site visit.
- The negative effect is strongest for forecasts issued soon after the visit and gradually attenuates toward zero by the end of the 30-day window.
- Table 5 tests whether pollution effects differ for high-pollution industries.

- The positive interaction between AQI and high-pollution status implies that the negative AQI effect is weaker for firms in high-pollution industries.

Detailed interpretation: Figure 1 is central to the affect interpretation: affective states should fade with time. Table 5 rejects a simple firm-information story: if analysts were inferring environmental liabilities from a polluted visit, the effect should be stronger for high-pollution firms, but the coefficient pattern is the opposite. These results move the paper from correlation to mechanism. They show that the pollution effect behaves like a temporary mood effect rather than a rational update about firm emissions.



effect of forecast delay. This figure shows how the coefficient estimates of *AQI* vary as a function of the numb

Figure 1: Forecast delay. The AQI coefficient is most negative for short windows and weakens as delay increases.

The effect of firm type.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the Air Quality Index of the visit city on the visit day, scaled by 1000. *HighPollution* is a dummy variable indicating that the visited firm belongs to one of the 16 high pollution industies defined by Ministry of Ecology and Environment of China (see text for details). Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature*, *Humidity*, *Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility*, *Turnover*, *Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)
Dependent variable	*Forecast_Optimism*	
HighPollution	0.344	−0.279
	(0.349)	(0.473)
AQI	−3.625**	−5.378***
	(1.461)	(1.604)
AQI * *HighPollution*		7.410**
		(2.992)
Year-quarter FEs	Yes	Yes
Day of week FEs	Yes	Yes
Industry FEs		
City FEs	Yes	Yes
Analyst FEs	Yes	Yes

Table 5: Firm type. Pollution-induced pessimism is not stronger for high-emission industries, undermining an emissions-information interpretation.

6.4 Table 6: The relation between air pollution and forecast optimism for different forecast horizons; Table 7: Air pollution adaptation and forecast optimism

Read:

- Table 6 interacts AQI with log forecast horizon. The interaction is negative and significant.
- Longer-horizon forecasts are more optimistic on average, but pollution reduces optimism more strongly for these forecasts.
- Table 7 compares site AQI to the analyst's home-city median AQI.
- The negative component is strongest when the site is more polluted than the analyst's home baseline, although the spline estimates are imprecise.

Detailed interpretation: Table 6 suggests that more subjective or uncertain forecasts are more vulnerable to affect. Table 7 is an adaptation test: if analysts accustomed to high pollution are less affected, the surprise component of pollution should matter more than the level alone. The evidence is suggestive rather than definitive. These tables deepen the behavioral interpretation by showing that the effect varies with forecast uncertainty and possibly with the analyst's normal pollution environment.

Table 6

The relation between air pollution and forecast optimism for different forecast horizons.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the (demeaned) Air Quality Index of the visit city on the visit date, scaled by 1000. log(*Horizon*) denotes the (demeaned) natural logarithm of the days elapsed between the forecast date and the corresponding date of the actual earnings announcement. Controls include *Hours_of_Sun*, *Temperature, Humidity, Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility, Turnover, Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)	(5)
Dependent variable			*Forecast_Optimism*		
AQI	-1.817^{*}	-2.285^{**}	-3.836^{***}	-4.388^{***}	
	(1.049)	(1.135)	(1.345)	(1.468)	
log(*Horizon*)	1.607^{***}	1.603^{***}	1.629^{***}	1.630^{***}	1.634^{***}
	(0.075)	(0.075)	(0.093)	(0.094)	(0.106)
log(*Horizon*)*AQI	-3.711^{***}	-3.676^{***}	-4.407^{***}	-4.337^{***}	-4.282^{***}
	(0.964)	(0.952)	(1.340)	(1.356)	(1.563)
Year-quarter FEs		Yes	Yes	Yes	
Day of week FEs		Yes	Yes	Yes	
Industry FEs			Yes	Yes	
City FEs			Yes	Yes	
Analyst FEs			Yes	Yes	
Controls				Yes	
Visit FEs					Yes
Observations	3824	3824	3824	3824	3824
R-squared	0.213	0.236	0.609	0.612	0.693

Table 6: Forecast horizon. Pollution effects are stronger for longer-horizon forecasts, where analyst judgment may be less anchored by precise information.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. ΔAQI equals to $AQI - AQI\_home$. *AQI* denotes the Air Quality Index of the site visit city on the visit date, scaled by 1000. *AQI_home* is the median AQI in the analyst's home city during the month preceding the site visit, scaled by 1000. Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature, Humidity, Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility, Turnover, Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)
Dependent variable		*Forecast_Optimism*		
ΔAQI(When $\Delta AQI \le 0$)	5.612^{**}	3.072	0.381	-0.395
	(2.593)	(2.472)	(4.719)	(4.529)
ΔAQI(When $\Delta AQI > 0$)	-5.035^{***}	-3.679^{***}	-4.993^{***}	-3.885^{**}
	(1.446)	(1.388)	(1.648)	(1.711)
Year-quarter FEs		Yes	Yes	Yes
Day of week FEs		Yes	Yes	Yes
Industry FEs			Yes	Yes
City FEs			Yes	Yes
Analyst FEs			Yes	Yes
Controls				Yes
Observations	3824	3824	3824	3824
R-squared	0.004	0.066	0.443	0.608

Table 7: Adaptation. Pollution relative to the analyst's home-city median tests whether familiar pollution exposure attenuates mood effects.

6.5 Table 8: Pollution, analyst skills, and forecasting bias; Table 9: The effect of analyst group visit on optimism

Read:

- Table 8 shows that the AQI coefficient remains negative after accounting for analyst star status, experience, and forecast accuracy.
- Experience is associated with less optimism, but skill variables do not strongly mitigate pollution's impact.
- Table 9 examines group visits.
- Having other analysts from different brokerages present attenuates the pollution effect: the AQI-by-other-brokerage interaction is positive and significant.
- Having other analysts from the same brokerage does not produce a similar debiasing pattern and may even amplify bias weakly.

Detailed interpretation: Table 8 indicates that expertise does not fully insulate analysts from affective environmental shocks. Table 9 suggests that independent social aggregation matters: outside-brokerage peers may discipline individual mood-driven judgment, while same-brokerage peers may be less independent. These are the paper's most direct tests of whether experience and social structure moderate behavioral bias. The evidence is consistent with debiasing through diverse external viewpoints.

Table 8
Pollution, analyst skills, and forecasting bias.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the Air Quality Index of the visit city on the visit day, scaled by 1000. *Star* is a dummy variable denoting whether the visiting analyst is ranked as a star by New Fortune magazine in the visit year. *Experience* is measured as the natural logarithm of the number of quarters since the analyst make his/her first forecast up to the end of the visit year. *Accuracy* is the average accuracy of the analyst's forecast within the past 1 year, with accuracy defined as the absolute difference between annual EPS forecast and realized EPS, scaled by the absolute value of realized EPS, multiplied by −1. Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature*, *Humidity*, *Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility*, *Turnover*, *Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable				*Forecast_Optimism*				
AQI	−3.770***	−3.746***	−4.005***	−4.020***	−4.075***	−3.570	−3.063	−3.028
	(1.423)	(1.423)	(1.458)	(1.460)	(1.480)	(2.870)	(2.583)	(3.534)
Star	0.073			0.140	−0.230			−0.259
	(0.273)			(0.275)	(0.388)			(0.379)
Experience		−0.295**		−0.317**		−0.287		−0.309
		(0.125)		(0.129)		(0.201)		(0.205)
Accuracy			0.080	0.079			0.049	0.044
			(0.066)	(0.065)			(0.109)	(0.108)
*AQI*Star*					3.624			4.857
					(3.123)			(3.070)
*AQI*Experience*						−0.089		−0.149
						(1.252)		(1.243)
*AQI*Accuracy*							0.364	0.430
							(0.782)	(0.776)
Year-quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of week FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyst FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3824	3824	3757	3757	3824	3824	3757	3757
R-squared	0.608	0.609	0.608	0.608	0.609	0.609	0.608	0.608

Table 8: Analyst skill. Skill and experience controls leave the negative AQI effect largely intact; interactions are weak and imprecise.

Table 9
The effect of analyst group visit on optimism.

Numbers in parentheses are standard errors clustered by firm. The sample covers the period from 2009 to 2015. The dependent variable in all columns is *Forecast_Optimism*, which denotes the difference between annual EPS forecast issued within calendar days [1,15] of the site visit and realized EPS, scaled by price as of the trading day prior to the forecast, multiplied by 100. *AQI* denotes the Air Quality Index of the visit city on the visit day, scaled by 1000. *GroupVisit_Same* is an indicator variable denoting that at least one other analyst from the same brokerage was present during the visit. *GroupVisit_Other* is an indicator variable denoting that at least one other analyst from a different brokerage was present during the visit. Controls include log(*Horizon*), *Hours_of_Sun*, *Temperature*, *Humidity*, *Precipitation*, *Wind_Speed*, log(*Assets*), *Market_to_Book*, *Intangible_Asset*, *Volatility*, *Turnover*, *Return*, *Analyst_Attention*, *Follow_Co_Num* and *Forecast_Num*, with output suppressed to conserve space. See the notes to Table 1 for detailed definitions of the control variables. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable			*Forecast_Optimism*			
AQI	−3.864***	−3.768***	−3.864***	−3.311**	−8.064***	−7.882***
	(1.432)	(1.420)	(1.432)	(1.525)	(2.447)	(2.459)
GroupVisit_Same	−0.277		−0.277	0.039		0.087
	(0.222)		(0.222)	(0.349)		(0.343)
GroupVisit_Other		0.019	0.017		−0.536*	−0.580*
		(0.166)	(0.165)		(0.315)	(0.316)
AQI ∗ *GroupVisit_Same*				−3.633		−4.684*
				(2.930)		(2.842)
AQI ∗ *GroupVisit_Other*					6.150**	6.752**
					(2.654)	(2.656)
Year-quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes
Day of week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes
City FEs	Yes	Yes	Yes	Yes	Yes	Yes
Analyst FEs	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3824	3824	3824	3824	3824	3824
R-squared	0.609	0.608	0.609	0.609	0.609	0.610

Table 9: Group visits. Cross-brokerage group visits attenuate the AQI effect, consistent with a wisdom-of-crowds or debiasing channel.

7 Short summary

- The paper studies whether air pollution affects sell-side analysts' earnings forecasts.
- The setting is Chinese corporate site visits from 2009 to 2015, matched to daily city-level AQI and forecast data.
- Forecast optimism is forecast EPS minus realized EPS, scaled by pre-forecast price and multiplied by 100.
- The baseline result is a robust negative relation between visit-day AQI and subsequent forecast optimism.
- The relation survives weather controls, firm controls, analyst controls, and fixed effects for year-quarter, day of week, industry, city, and analyst.
- AQI-category estimates show a dose-response pattern.
- Lagged and future AQI do not explain the result, supporting a visit-day exposure interpretation.
- The effect fades as the delay between the visit and the forecast increases.
- The effect is not stronger for high-pollution firms, which weakens the hypothesis that analysts are learning negative fundamentals from firm emissions.
- The effect is stronger for longer-horizon forecasts, where uncertainty and subjective judgment are greater.
- Analyst experience and ability do not strongly eliminate the bias.
- Cross-brokerage group visits attenuate the effect, suggesting that diverse social interaction can reduce mood-driven bias.
- The paper's main contribution is to show that expert forecasts in capital markets are affected by transitory environmental conditions in a way consistent with affect.

8 Conclusion

8.1 Core conclusion

Dong, Fisman, Wang, and Xu show that air pollution during corporate site visits predicts lower subsequent earnings forecast optimism by professional financial analysts. The effect remains after controlling for weather, firm characteristics, analyst characteristics, and multiple fixed effects. The result suggests that even expert information intermediaries are not immune to transitory environmental influences on affect and judgment.

8.2 Why the conclusion is economically important

The finding matters because analysts play a central role in capital markets. Their forecasts affect investor expectations, information dissemination, and potentially asset prices. If irrelevant environmental conditions can shift analyst forecasts, then information production in financial markets is not purely a function of data quality, incentives, and expertise. It is also shaped by the human conditions under which information is processed.

8.3 Mechanism and interpretation

The most persuasive evidence for the affect channel is the combination of four patterns. First, the effect is strongest shortly after the visit and fades with time. Second, lagged and future pollution do not explain the forecast change. Third, the effect is weaker for high-pollution firms, which undermines a rational inference about environmental liabilities. Fourth, cross-brokerage group visits attenuate the effect, suggesting that independent discussion can debias individual mood-driven assessments.

8.4 Limitations

The paper does not randomly assign analysts to polluted days. The estimates rely on conditional variation in AQI around site visits. It is therefore possible that some unobserved city-day conditions correlated with pollution affect analyst beliefs. The paper addresses this with weather controls, fixed effects, timing tests, and heterogeneity tests, but the design is best interpreted as strong quasi-experimental evidence rather than a fully randomized experiment.

8.5 Takeaway

The paper's broad takeaway is that expert judgment in finance is environmentally situated. Air pollution can affect professional forecasts, but the effect is temporary and can be mitigated by information aggregation across independent analysts. This makes the paper important for behavioral finance, environmental economics, and the design of information-production institutions.